## ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** |
Class 7 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 10,00,000 | Multiply and place commas. |
| 2 | **(a)** 36 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 3 | **(b)** 1 | Count the total number of decimal places. |
| 4 | **(a)** 19 | Substitute both values. |
| 5 | **(b)** 1250000 | Divide. |
| 6 | **(b)** 27 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 7 | **(d)** 1.2 | Shift the decimal points of both numbers equally. |
| 8 | **(a)** $4x + 7y$ | Collect the like terms separately. |
| 9 | **(b)** 10000 | Divide $10^7$. |
| 10 | **(c)** 50 | Evaluate first. |
| 11 | **(b)** 11.75 | Multiply first, then add. |
| 12 | **(b)** 23 | Work out the bracket first. |
| 13 | **(a)** 19,99,999 | 999999 + 1000000. |
| 14 | **(b)** 9 | Divide the product by the multiplier, then subtract. |
| 15 | **(a)** 1.25 | Divide the total by the number of persons. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Multiply and add commas (1); $35,10,000$ (1). | 1+1 |
| 17 | Innermost bracket first; Then the next bracket (1); Then the rest in order; Value $= 18$ (1). | 1+1 |
| 18 | $5.6 + 3.45 = 9.05$ (1); $9.05 - 2.8 = 6.25$ (1). | 1+1 |
| 19 | Expand the bracket: $3x + 12$ (1); Add $5x$: $8x + 12$ (1). | 1+1 |
| 20 | Divide step by step (1); $36,00,000$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | In 8 years: $1,25,000 \times 8 = 10,00,000$ km (1); Trains needed $= 60,00,000 \div 10,00,000 = 6$ (2). | 1+2 |
| 22 | Cost $= 50 \times 40 = 2000$; Sales $= 30 \times 60 + (50 - 30) \times 35 = 2500$ (1); Result $= 500$ (1); Profit (1). | 1+1+1 |
| 23 | Distance $= 12.5 \times 8 = 100$ km (1); Litres $= 75 \div 12.5 = 6$ (2). | 1+2 |
| 24 | Sum $= 10n + 3$ (1); Difference $= 4n + 7$ (1); At $n = 4$: sum $= 43$ and difference $= 23$ (1). | 1+1+1 |
| | **OR:** (i) $34,57,000$ (1); (ii) $35,00,000$ (1); Difference $= 43,000$ (1). | 1+1+1 |
| 25 | Brackets first; Multiplication (1); Final answer (1); Value $= 180$ (1). | 1+1+1 |
| | **OR:** Cost $= 8 \times 7.5 = 60$ (1); Balance $= 100 - 60 = 40$ (2). | 1+2 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | 2nd year $= 14,00,000$ (1); 3rd year $= 15,50,000$ (1); 4th year $= 17,00,000$ (1); Total $= 59,00,000$ (2). | 1+1+1+2 |
| | **OR:** Chairs after discount $= 40 \times (450 - 50) = 16000$ (1); Tables $= 10 \times 1200 = 12000$ (1); Total $= 28000$ (1); Expression: $40 \times (450 - 50) + 10 \times 1200$ (2). | 1+1+1+2 |
| 27 | $9.5 \div 1.2$ gives 7 whole pieces (1); Length used $= 7 \times 1.2 = 8.4$ m (2); Left over $= 9.5 - 8.4 = 1.1$ m (2). | 1+2+2 |
| | **OR:** Sales $= 48x + 36y$ (1); At the given values: $480 + 540$ (2); Total sales $= ₹1020$ (2). | 1+2+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** $5x + 7$ | Subtract the $x$-terms and the constants separately. |
| 2 | **(b)** 1800000 | Round first. |
| 3 | **(b)** 325 | Total minus reserved. |
| 4 | **(d)** 12 | Multiply first, then subtract. |
| 5 | **(c)** 43 | Substitute the value of $n$. |
| 6 | **(c)** 86400 | 1 hour = 3600 seconds. |
| 7 | **(c)** $(8 + 4) \times 3$ | Compare the two groupings. |
| 8 | **(d)** 3.45 | 1 m = 100 cm, so write the smaller unit as a decimal part. |
| 9 | **(d)** $3x + 5y$ | Multiply each price by its quantity. |
| 10 | **(d)** 3650 | Multiply. |
| 11 | **(a)** 15 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 12 | **(a)** 3.46 | Look at the thousandths digit. |
| 13 | **(d)** 15 | $a^2 = a \times a$. |
| 14 | **(d)** 34,50,000 | Multiply and place commas. |
| 15 | **(c)** 28 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $6 \times (9 + 8)$ (1); Value $= 102$ (1). | 1+1 |
| 17 | $1.2 \times 0.5 = 0.6$ (1); $0.6 \div 0.04 = 15$ (1). | 1+1 |
| 18 | $(9x + 2) - (3x - 5)$ (1); Answer: $6x + 7$ (1). | 1+1 |
| 19 | 1 day = 24 × 60 = 1440 minutes (1); $30 \times 1440 = 43,200$ (1). | 1+1 |
| 20 | First $= 60$; Second $= 60$ (1); Equal (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | For 9 triangles: $2(9) + 1 = 19$ (1); For 14 triangles: $2(14) + 1 = 29$ (2). | 1+2 |
| 22 | Exact $= 45,67,900$ (1); Estimate $= 46,00,000$ (1); The estimate is close to the exact answer (1). | 1+1+1 |
| 23 | Try bracketing different parts (1); Correct: $(2 + 3) \times 4 + 5$ (1); Check value 25 (1). | 1+1+1 |
| 24 | Compare whole parts, then tenths, then hundredths (1); Descending order: $4.55, 4.5, 4.05, 4.005$ (1); Difference $= 4.55 - 4.005 = 0.545$ (1). | 1+1+1 |
| | **OR:** Fare $= 45 + 11k$ (1); For 8 km: $45 + 11(8) = 133$ (1); For 13 km: $45 + 11(13) = 188$ (1). | 1+1+1 |
| 25 | In 12 years: $84,000 \times 12 = 10,08,000$ km (1); Trains needed $= 50,40,000 \div 10,08,000 = 5$ (2). | 1+2 |
| | **OR:** Cost $= 40 \times 55 = 2200$; Sales $= 25 \times 70 + (40 - 25) \times 45 = 2425$ (1); Result $= 225$ (1); Profit (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Days $= \lfloor 12,50,000 \div 40,000 \rfloor = 31$ full days (1); Remaining $= 10,000$ m (2); Total days to finish: 32 (2). | 1+2+2 |
| | **OR:** Evaluate brackets first (1); Then multiplication/division (1); Value $= 465$ (1); A suitable story is consistent with the operations (2). | 1+1+1+2 |
| 27 | Time $= 90 \div 7.5 = 12$ minutes (1); After 6 minutes: $7.5 \times 6 = 45$ litres (2); More needed $= 90 - 45 = 45$ litres (2). | 1+2+2 |
| | **OR:** Mother now $= 3r + 4$ (1); After 5 years: Raju $= r + 5$, mother $= 3r + 9$ (1); Sum $= 4r + 14$ (1); At $r = 10$: 54 years (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 0.37 | A number between them is greater than the smaller and less than the larger. |
| 2 | **(a)** 11 | Work out the numerator, then divide. |
| 3 | **(c)** 4800000 | Divide. |
| 4 | **(a)** 69 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 5 | **(c)** 1.2 | Count the total number of decimal places. |
| 6 | **(d)** 30 | Substitute both values. |
| 7 | **(c)** 100 | Divide $10^7$. |
| 8 | **(c)** 22 | Evaluate first. |
| 9 | **(a)** 0.55 | Shift the decimal points of both numbers equally. |
| 10 | **(d)** $8x + 9y$ | Collect the like terms separately. |
| 11 | **(a)** 1 | Consecutive numbers. |
| 12 | **(b)** 8 | Divide the product by the multiplier, then subtract. |
| 13 | **(c)** 2.5 | Multiply first, then add. |
| 14 | **(c)** 31 | Work out the bracket first. |
| 15 | **(d)** 2000000 | Round first. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Look at the digit after the place being rounded; (i) $8.7$ (1); (ii) $8.75$ (1). | 1+1 |
| 17 | Substitute: $5^2 + 3(5) - 4$ (1); $= 36$ (1). | 1+1 |
| 18 | Multiply and add commas (1); $22,68,000$ (1). | 1+1 |
| 19 | Innermost bracket first; Then the next bracket (1); Then the rest in order; Value $= 46$ (1). | 1+1 |
| 20 | $12.05 + 7.9 = 19.95$ (1); $19.95 - 6.35 = 13.6$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Distance $= 15.5 \times 6 = 93$ km (1); Litres $= 62 \div 15.5 = 4$ (2). | 1+2 |
| 22 | Sum $= 14n + 3$ (1); Difference $= 4n + 5$ (1); At $n = 3$: sum $= 45$ and difference $= 17$ (1). | 1+1+1 |
| 23 | (i) $56,78,000$ (1); (ii) $57,00,000$ (1); Difference $= 22,000$ (1). | 1+1+1 |
| 24 | Brackets first; Multiplication (1); Final answer (1); Value $= 120$ (1). | 1+1+1 |
| | **OR:** Cost $= 6 \times 12.25 = 73.5$ (1); Balance $= 100 - 73.5 = 26.5$ (2). | 1+2 |
| 25 | For 12 triangles: $2(12) + 1 = 25$ (1); For 20 triangles: $2(20) + 1 = 41$ (2). | 1+2 |
| | **OR:** Exact $= 76,54,321$ (1); Estimate $= 77,00,000$ (1); The estimate is close to the exact answer (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | 2nd year $= 26,50,000$ (1); 3rd year $= 29,00,000$ (1); 4th year $= 31,50,000$ (1); Total $= 1,11,00,000$ (2). | 1+1+1+2 |
| | **OR:** Chairs after discount $= 30 \times (600 - 100) = 15000$ (1); Tables $= 8 \times 1500 = 12000$ (1); Total $= 27000$ (1); Expression: $30 \times (600 - 100) + 8 \times 1500$ (2). | 1+1+1+2 |
| 27 | $7.8 \div 0.9$ gives 8 whole pieces (1); Length used $= 8 \times 0.9 = 7.2$ m (2); Left over $= 7.8 - 7.2 = 0.6$ m (2). | 1+2+2 |
| | **OR:** Sales $= 45x + 30y$ (1); At the given values: $900 + 300$ (2); Total sales $=$ ₹1200 (2). | 1+2+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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